

1. Explain Cyber Deception

Cyber deception is a cybersecurity strategy that uses fake systems, files, credentials, or network services to mislead attackers and detect malicious activity. Instead of only blocking attacks, cyber deception creates realistic decoys that appear valuable to attackers. When an attacker interacts with these decoys, security teams receive alerts and can study the attack without putting real systems at risk.

The main goal of cyber deception is to detect threats early, delay attackers, collect intelligence about their methods, and improve an organization's overall security posture. Since legitimate users have no reason to access these fake resources, any interaction is considered highly suspicious.

2. Research Honeypots

A **honeypot** is a decoy computer system or service designed to attract cyber attackers. It looks like a real system but does not contain important business data. Instead, it is monitored to observe attacker behavior and collect threat intelligence.

Types of Honeypots

a) Low-Interaction Honeypot

- Simulates only a few services.
- Easy to deploy and maintain.
- Used mainly to detect basic attacks.

b) High-Interaction Honeypot

- Mimics a real operating system and applications.
- Allows attackers to interact more freely.
- Provides detailed information about attacker techniques but requires stronger security controls.

Benefits of Honeypots

- Detect unauthorized access attempts.
- Collect information about new attack techniques.
- Reduce false positive alerts.
- Improve incident response.

- Support security research and threat intelligence.

Limitations

- Attackers may recognize the honeypot.
- It protects only if attackers interact with it.
- High-interaction honeypots require careful monitoring and maintenance.

3. Analyze an Attacker Interaction Scenario

Scenario

A company deploys a honeypot server that appears to be a financial database. The server is intentionally placed on the internal network with fake user accounts and sample financial records. One evening, the Security Operations Center (SOC) receives an alert showing multiple failed login attempts followed by a successful login to the honeypot. The attacker begins exploring folders and attempting to download files.

Analysis

Since the server is a honeypot, no legitimate employee should access it. The login activity immediately indicates suspicious behavior. Security analysts begin monitoring the attacker's actions while ensuring the attacker remains isolated from the organization's production systems.

During the investigation, analysts discover that the attacker used stolen credentials to gain access. The attacker attempted to search for sensitive financial data, but only fake information was available. Meanwhile, the SOC collected valuable information, including the attacker's IP address, commands executed, login times, and techniques used during the attack.

Indicators Observed

- Multiple failed login attempts.
- Successful login using suspicious credentials.
- Access to a honeypot server.
- Attempts to browse sensitive directories.
- Unusual outbound network connections.

Response Steps

- Continue monitoring the attacker without revealing the deception.
- Collect logs and forensic evidence.
- Block the attacker's IP address if necessary.
- Reset any compromised user credentials.
- Scan the production network for similar malicious activity.
- Update security rules based on the attacker's techniques.
- Document the incident for future reference.